

Assignment 03

Task 1 : Time Value of Money

Given :

$$CF_4 = 1000$$

$$t = 4$$

$$r = 8\% \Rightarrow 0.08$$

Formulas

$$1) DF_t = \frac{1}{(1+r)^t}$$

$$2) PV = CF_t \times DF_t$$

Solution :

$$DF_4 = \frac{1}{(1+0.08)^4} \Rightarrow \frac{1}{1.3604} \Rightarrow 0.7350$$

$$PV = 1000 \times 0.7350$$

$$= 735.029 \approx 735.03 \text{ Rs}$$

Hence, 1000 Rs received after 4 years will be worth of 735.029 Rs at a discount rate of 8% today's.

Task 2 : Net Present Value (NPV)

Given :

$$C_0 = -10,000$$

$$C_1 = 2000$$

$$C_2 = 3000$$

$$C_3 = 4000$$

$$C_4 = 4000$$

$$r = 10\% \Rightarrow 0.1$$

Formulas :

$$1) NPV = \sum_{t=0}^n \frac{C_t}{(1+r)^t}$$

2) if $NPV > 0$ accepted
else ($NPV \leq 0$) rejected

Solution:

$$NPV = \frac{C_0}{(1+r)^0} + \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3} + \frac{C_4}{(1+r)^4}$$

$\rightarrow (i)$

$$\frac{C_0}{(1+0.1)^0} \Rightarrow \frac{-10,000}{(1+0.1)^0} \Rightarrow -10,000 \text{ (cash out flow)}$$

$$\frac{C_1}{(1+0.1)^1} \Rightarrow \frac{2000}{(1+0.1)^1} \Rightarrow 1818.1818 \text{ (cash inflow)}$$

$$\frac{C_2}{(1+0.1)^2} \Rightarrow \frac{3000}{(1+0.1)^2} \Rightarrow 2479.3388 \text{ (inflow)}$$

$$\frac{C_3}{(1+0.1)^3} \Rightarrow \frac{4000}{(1+0.1)^3} \Rightarrow 3005.2592 \text{ (inflow)}$$

$$\frac{C_4}{(1+0.1)^4} = \frac{4000}{(1+0.1)^4} \Rightarrow 2732.0538 \text{ (inflow)}$$

put all values in eq. (i)

$$NPV = -10,000 + 1818.1818 + 2479.3388 + 3005.2592 + 2732.0538$$

$$NPV = 34.8336$$

as the $NPV > 0$ you it should be accepted. Also the present value of inflow is also larger i.e.,

$$1034.8336 > 10,000$$

Task 3 : Payback Period

Given :

Initial Investment = 6000

$$CF_1 = 1000$$

$$CF_2 = 1500$$

$$CF_3 = 2500$$

$$CF_4 = 3000$$

To find :

1) Cumulative Cash flow

2) Payback Period

Solution :

$$\bullet \text{ Cumulative } CF_0 \text{ (at time of investment)} = -6000$$

$$\bullet \text{ Cumulative } CF_1 = -6000 + 1000 \\ = -5000$$

$$\bullet \text{ Cumulative } CF_2 = -5000 + 1500 \\ = -3500$$

$$\bullet \text{ Cumulative } CF_3 = -3500 + 2500 \\ = -1000$$

$$\bullet \text{ Cumulative } CF_4 = -1000 + 3000 \\ = +2000$$

Pay back Year = ?

Pay back occur during year 4

$$\text{Fraction of Year 4 required} = \frac{1000}{3000} \Rightarrow 0.3333$$

$$\text{in months} = 0.3333 \times 12 \Rightarrow 4$$

So, payback period is 3 years and 4 months.

Task 4 : Internal Rate of Return (IRR)**Given :**

initial investment = 5000 Rs

Annual receipts = 2000 Rs

for 3 years

Formula :

$$0 = \sum_{t=1}^3 \frac{2000}{(1+IRR)^t} - 5000$$

Solution :

$$0 = -5000 + \frac{2000}{(1+IRR)^1} + \frac{2000}{(1+IRR)^2} + \frac{2000}{(1+IRR)^3}$$

Trial Method :

$$\text{NPV}(0\%) = -5000 + 2000 + 2000 + 2000$$

$$= +1000$$

$$\text{NPV}(5\%) = -5000 + \frac{2000}{(1+0.05)^1} + \frac{2000}{(1+0.05)^2}$$

$$+ \frac{2000}{(1+0.05)^3} \Rightarrow +446.4960$$

$$\text{NPV}(10\%) = -5000 + \frac{2000}{(1.10)^1} + \frac{2000}{(1.10)^2} + \frac{2000}{(1.10)^3}$$

$$= -26.2960$$

∴ This shows IRR lies between 5% and 10%

Approximate IRR %

$$\begin{aligned}
 IRR &= r_1 + \frac{NPV(r_1)}{NPV(r_1) - NPV(r_2)} (r_2 - r_1) \\
 &= 0.05 + \frac{446.4960}{446.4960 - (-26.2960)} (0.10 - 0.05) \\
 &= 0.05 + 0.94438 (0.10 - 0.05) \\
 &= 0.05 + 0.47219 \\
 &= 0.0972 \approx 9.72\%
 \end{aligned}$$

Hence the IRR that makes $NPV = 0$ is approximately 9.72% if rate of return is less than 9.72% accept otherwise reject.

Task 5: Sensitivity Analysis

Given :

initial development cost = 20,000

selling price per unit = 5000

Expected sale :

Year 1 = 30 units

2 = 35 units

3 = 40 units

Discount rate (r) = 10% $\Rightarrow$ 0.10

sales drop at Year 3 = 40 $\rightarrow$ 20 units

Formula :

$$NPV = \sum_{t=0}^n \frac{C_t}{(1+r)^t}$$

Solution : Predicted Case :

$$C_0 = -20,000$$

$$C_1 = \frac{30 \times 5000}{(1+0.10)^1} = 1,363,636.364$$

$$C_2 = \frac{35 \times 5000}{(1+0.10)^2} = 1,446,280.992$$

$$C_3 = \frac{40 \times 5000}{(1+0.10)^3} = 1,502,629.602$$

$$\begin{aligned} NPV_{\text{base}} &= -20,000 + 1,363,636.364 \\ &\quad + 1,446,280.992 + 1,502,629.602 \\ &= -20,000 + 4,312,546.958 \\ &= 4,112,546.958 \end{aligned}$$

Sensitivity Case :-

$$C_3 = \frac{20 \times 5000}{(1+0.10)^3} \Rightarrow 751,314.8009$$

$$\begin{aligned} NPV_{\text{new}} &= -20,000 + 1,363,636.364 \\ &\quad + 1,446,280.992 + 751,314.8009 \\ &= 3,361,232.2157 \end{aligned}$$

Comparison

$$\begin{aligned} \Delta NPV &= NPV_{\text{base}} - NPV_{\text{new}} \\ &= 4,112,546.958 - 3,361,232.2157 \\ &= 751,314.7423 \end{aligned}$$

Hence the project is robust to this assumption (Year 3 sales drop 40 → 20 units) even with 50% drop in Year 3 sales as NPV remains positive

Task 6: Lease vs Buy Decision

Given:

buy = £8995

lease = £500 down payment
plus £400 per month
for 24 months

monthly discount rate $r = 0.2466\%$

Formula:

$$PV = \sum_{t=1}^n \frac{400}{(1+r)^t} + 500$$

Solution:

$$t_0 = 500$$

$$t_1 = \frac{400}{(1+0.2466\%)^1} \Rightarrow \frac{400}{(1+0.002466)^1} \Rightarrow 399.016$$

in the same way we will be calculating all 24 values

$$t_2 = 398.0344$$

$$t_3 = 397.0553$$

$$t_4 = 396.0786$$

$$t_5 = 395.1042$$

$$t_6 = 394.1323$$

$$t_7 = 393.1498$$

$$t_{16} = 384.3456$$

$$t_8 = 392.1329$$

$$t_{17} = 383.7642$$

$$t_9 = 391.1038$$

$$t_{18} = 382.8239$$

$$t_{10} = 390.1444$$

$$t_{19} = 381.7126$$

$$t_{11} = 389.1789$$

$$t_{20} = 380.7737$$

$$t_{12} = 388.1233$$

$$t_{21} = 379.8370$$

$$t_{13} = 387.9026$$

$$t_{22} = 378.9026$$

$$t_{14} = 386.1478$$

$$t_{23} = 377.9705$$

$$t_{15} = 385.1111$$

$$t_{24} = 377.0407$$

$$PV = \frac{39}{t_0} t_0 + t_1 + t_2 + \dots + t_{24}$$

$$PV = 39500 + 399.016 + 398.0344 + \dots + 377.9705 + 377.0407$$

$$PV = 9810.3005$$

$$PV_{\text{lease}} \approx 9810.30$$

comparison:

$$\Delta PV = PV_{\text{lease}} - PV_{\text{buy}}$$

$$\approx 9810.3005 - 8995$$

$$\approx 815.3$$

Hence buying saves approximately £815.3 so it's better to buy outright for £8995 than to get it on lease (£9810.30)

Task 1 : contract writing.

• Title :

Vehicle Purchase Agreement

• Parties :

Party A (Buyer) : Arshah

Party B (seller) : Lamees

• Purpose :

The Party A will be using car for operational purpose. The following two alternatives were being analysed :

1. Buy the car with one time payment of £ 8995
2. Buy the car on lease. £500 payment at the start later £400 per month for 24 months.

The discount rate of 0.2466% will be applied annually.

• Financial Analysis :

By using formula :

$$PV_{\text{lease}} = \sum_{t=1}^{24} \frac{400}{(1+r)^t} + 500$$

The lease value has calculated which is £9810.3005 while the one time payment is £ 8995. if we calculate the difference :

$$\begin{aligned}\Delta PV &= PV_{\text{lease}} - PV_{\text{buy}} \\ &= 9810.3005 - 8995 \approx £815.3\end{aligned}$$

The car become costlier of about £815.3 if purchased on lease.

• Decision Clause :

Based on financial evaluation Party A shall purchase the vehicle outright for £8995. It's more approximately £815.3 relative to lease.

• Risk clause :

if market interest rate increase by more than 2% per annum. then both parties should re-evaluate this decision within 30 days.

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Signatures

Party

Signature

Date

Party A (Buyer)

et al

4th October 2025

Party B (seller)

Lammas

4th October 2025